MILITARY SPECIFICATION

PROCESSING MACHINE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of field processing machine, with and without accessories, that is suitable for 10-second to 1-minute processing of radiographic paper and developing assemblies.

1.2 Classification.

1.2.1 Classes. The processing machine shall be of one type and of the following classes:

Class I - Processing machine and accessories.
Class II - Processing machine only.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 South 20th Street, Philadelphia, Pa. 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6525

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SPECIFICATIONS

FEDERAL

GG-S-176	- Screen, X-ray Intensifying.
NN-P-530	- Plywood, Flat Panel.
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet.
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-A-601	- Aluminum Alloy Sand Castings.
TT-E-529	- Enamel, Alkyd, Semi-Gloss.
TT-P-1757	- Primer Coating, Zinc Chromate, Low Moisture Sensitivity.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

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MIL-P-116	- Preservation-packaging, Methods Of.
MIL-T-152	- Treatment, Moisture-and-Fungus-Resistant, Of Communications, Electronic, and Associated Electrical Equipment).
MIL-V-173	- Varnish, Moisture & Fungus Resistant, (For the Treatment of Communication, Electronic, and Associated Electrical Equipment).
MIL-M-7298	- Manual, Technical: Commercial Equipment.
MIL-F-10336	- Fiber Sheet, Vulcanized.

STANDARDS

FEDERAL

FED-STD-66	- Steel, Chemical Composition and Hardenability.
FED-TEST METHOD-STD-151	- Metals, Test Methods.
FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

* 3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

* 3.2 Material. All parts entering into the construction of the processing equipment shall be manufactured from materials which are entirely satisfactory for the purpose and of sufficient strength to maintain operational alignment for the life of the equipment. The equipment shall be resistant to damage by fungus or other tropical conditions.

* 3.2.1 Cassettes. The aluminum sheet for mounting the intensifying screen shall be alloy 3003-H14 conforming to QQ-A-250/2. The cassette frame shall be cast from an aluminum alloy conforming to QQ-A-591 or QQ-A-601. Cassette hardware and rivets shall be constructed of corrosion-resisting steel, AISI 300 or 400 series of FED-STD-66. As an alternate, the rivets and washers used to secure the pressure springs to the rear of the cassette and the rivets that are used to secure the nameplate shall be brass. Other cassette parts shall be constructed as specified herein.

3.2.1.1 Adhesive. Adhesive for mounting the intensifying screen and fabrics in the cassette shall not be injurious to these parts or to the film. The adhesives shall leave no shadows on the exposed film.

3.2.2 Carrying case. Plywood shall be 3-ply, gum or fir conforming to type I of NN-P-530, 3/8 inch thick throughout for both the body and cover sections. Vulcanized fiber shall conform to MIL-F-10336. The fiber shall have a smooth, calendered, homogenous, flat surface. The color of the fiber shall extend through the sheet, and shall conform to color 24087 of FED-STD-595 (see 6.5). All carrying case hardware used in the manufacture of the carrying case shall be of the standard commercial type, stamped or formed from cold-rolled sheet steel. Dimensions and shape shall be adequate for the purpose intended. Rivets and washers (burs) shall be of the size and shape suitable for the intended purpose. Rivets shall be fabricated from rivet-grade steel. Riveting washers shall be fabricated from carbon steel.

* 3.2.3 Tolerances. Unless otherwise specified, commercial tolerances shall apply for metal thicknesses.

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* 3.3 Design and construction. The class I processing machine shall consist of a processing machine, three cassettes, a carrying case, tools, and repair parts (see 3.4). The class II processing machine shall consist of the processing machine only. The processing machine and a cassette shall dry-process exposed radiographic paper and developer assemblies without the need for darkroom facilities.

* 3.3.1 Processing machine. The processing machine shall be designed to accommodate cassettes (see 3.3.2), and shall provide a light-tight chamber which, in the operating position, shall prevent premature exposure of the film packet during the processing. The apparatus shall be pushbutton controlled and motor operated with timing and interlock devices to insure proper and safe processing of the film packet. The processing mechanism shall be mounted integrally upon its own frame and shall be secured within the light-tight chamber. The processing mechanism shall include a shelf designed to fit the cassette without binding. Two rollers shall be mounted through a linkage to the pressure bar. Each roller shall be constructed of an aluminum mandrel covered with a vulcanized rubber coating. The rollers shall be precision ground to close tolerances. The linkage of the upper roller shall terminate in adjustable spring mountings to provide the proper pressure of the roller upon the film packet. One pair of adjustable spring mountings shall provide the proper pressure for the rupture of the pod(s) containing the developing agent, and for the squeegeeing of the developing agent in the developer packet. The linkage of the pressure bar shall have adjusting screws to provide for parallel contact adjustment of the rubber rollers. The lower roller shall be driven by means of a motor through a gear train. The motor shall operate satisfactorily over the range of 90 to 120 volts and 50 to 60 Hz, the film shall exhibit normal development and the machine shall operate properly when tested as specified in 4.4.1.2.

3.3.1.1 Cassette safety switch. A cassette safety switch shall be incorporated which shall break the circuit when the cassette is in place and its pressure release bar has not been raised to permit the film packet to be withdrawn. This switch shall be operated by an arm extending over the cassette shelf. This arm shall be pivotally mounted in the frame, and shall have affixed to it a flag designed to strike the cassette pressure lever. This arm shall terminate in a slotted bracket mounted to the inside of the lid of the chamber. This arm shall act further to restrict the opening of the lid of the chamber to approximately 100°.

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3.3.1.2 Film switch. A normally closed film switch shall be mounted on the cassette shelf near the rolls. This switch shall have a pressure finger which shall be depressed by the film packet to de-energize the pressure latch solenoid, and then energize the solenoid when the film has passed the switch. This pressure finger shall project through a stainless steel shelf attached to the base casting immediately adjacent to a lower rubber roller, and the shelf shall act as a stop for the cassette and a guide for the film packet.

3.3.1.3 Roller-pressure mechanism. The pressure bars of the upper roller shall be operated through the adjustable springs (see 3.3.1) through a pair of linkage bars tied together with a heavy torsion bar. A thumb lever, mounted upon a second torsion shaft with fingers bearing against the torsion bar, shall raise the torsion bar equally at both sides to provide equal pressure to the upper roller linkage bars. A solenoid-operated mechanism shall latch the thumb lever to maintain the pressure between the rollers. The solenoid shall be energized by the film packet switch to release the pressure on the rollers 1/4 inch or slightly less before the end of the film packet is reached, and to return the mechanism simultaneously to the unlatched position. This shall prevent damage to the rollers. Immediately adjacent to the thumb lever there shall be a release lever designed to unlatch the solenoid mechanically, if an error has been made in the insertion of the film packet tab between the rollers preliminary to processing. This thumb lever shall be so positioned that the metallic channel on the imbibition chamber shall depress it, when the imbibition chamber is stored within the processor, to preclude the possibility of storing the processor with the rubber rollers in contact.

* 3.3.1.4 Bearings. All rotating or oscillating shafts shall operate within self lubricating bronze bushings (see 6.7) with the exception of the main rubber-coated pressure rollers, which shall be mounted in high-grade ball or roller type bearings.

3.3.1.5 Switches. All electrical switches shall be identical to simplify replacement. They shall also be adjustable to decrease or increase sensitivity as required for proper operation.

3.3.1.6 Leaf spring. A heavy leaf spring shall be mounted beneath the cassette shelf, shall bear against the pressure bar, and shall act to separate the rollers immediately after the solenoid is energized.

3.3.1.7 Electrical components. All electrical components, including the roller-driving motor, synchronous timer, operating switches, solenoids, and latch mechanism, shall be mounted upon an integral base and attached to the side of the processor base by means of three screws. This shall be readily removable for ease of servicing or parts replacement.

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* 3.3.1.8 Pushbutton. A light-tight pushbutton shall be mounted in the frame of the unit. This pushbutton shall serve to energize the roller-driving motor. The pushbutton shall also energize a motor-driven timer of the adjustable type having a range of 0 to 57 seconds, minimum. The timer shall activate a buzzer at the expiration of the preset time.

3.3.1.9 Imbibition chamber. A separate light-tight imbibition chamber shall be provided, mounted by a sliding fit between the female channel on the imbibition chamber and the male fitting on the processor housing. A plunger-type catch shall retain the imbibition chamber on the housing. This chamber shall serve to receive the film packet as it leaves the rollers and store it during the developing interval. This imbibition chamber shall be removable from the main chamber and stored within it during transport.

* 3.3.1.10 Line cable. The unit shall include a fuse of suitable capacity which is accessible beneath the mechanism cover. The unit shall be provided with a line cable, 20 feet in length overall, of three-conductor service cord, SJO, or SJ with No. 18 conductors. The mechanism end of this cable shall be provided with a cord-mounted electrical 3-prong plug conforming to UL hospital grade standards.

3.3.1.11 Processor mechanism chamber. The processor mechanism shall be housed in a chamber having overall dimensions of 22' by 18-1/4 by 5-1/2 inches with tolerances of + 1/16 inch. This chamber shall be constructed of drawn 0.047 inch thick, cold-rolled steel. The lid of the chamber shall have around the periphery of its opening, a recess in which a sponge-rubber gasket is inserted to render the unit light-tight. The lower half of the chamber shall have a bead around the periphery of its opening to provide a rounded surface to bear against the sponge-rubber gasket in the periphery of the lid. A corrosion-resisting steel continuous hinge shall be welded and bolted along one edge of the chamber. The end of the chamber adjacent to the rollers of the mechanism shall have a aperture pierced within it, and shall have a fitting to accommodate the imbibition chamber. Four 3/8 inch bosses shall be depressed in the bottom of the chamber to elevate the bottom sufficiently to clear all retaining screws. A latch shall be provided on the side of the chamber opposite the hinge, which shall securely latch the lid to the bottom when in the closed position.

3.3.1.12 Moisture and fungus resistance. Electrical parts shall be moisture and fungus resistant treated in accordance with MIL-T-152. As an alternate, one of the treatments described in 6.11 shall be acceptable.

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3.3.2 Cassettes. The cassettes shall be designed to accommodate 9-3/8 inch by 10-inch radiographic paper and developer assemblies (see 6.1), and shall operate with the processing machine described herein to dry-process the exposed film packet. All loading and processing shall be possible in daylight without the use of a darkroom. The cassette dimensions shall be such as to permit use in the cassette tray of conventional Bucky diaphragms or cassette changers with proper adapters. Cassettes shall be provided with an adequate seal against light to permit loading without resorting to the use of a darkroom.

* 3.3.2.1 Frame. The unit shall include a hard, chrome plated aluminum frame designed to accommodate the packet which shall have an image size of approximately 9-3/8 by 10 inches. The exterior dimensions of the cassette shall be 13-15/16 inches wide by 11-3/8 inches long with tolerances of ± 1/32 inch. The cassette thickness shall be 21/32 ± 1/64 inch. These dimensions shall permit the cassette to be used in place of a 10 by 12 inch radiographic cassette for general bucky work. The closed cassettes shall have no protruding parts that interfere with stacking or with their insertion or withdrawal from cassette holding devices. The cassette surfaces shall be flat and square within 1/32 inch. The thickness of the frame at the ends of the cassette shall be 1/4 ± 1/32 inch.

* 3.3.2.2 Front. The cassette shall have a bakelite front, designed to float within the frame to minimize distortion due to expansion and contraction. When tested as specified in 4.4.3, the front shall have an opacity to x-rays not greater than the equivalent to .25 millimeter (mm.) of aluminum at 60 kilovolts peak, and shall be free of voids, imperfections, and included foreign matter which would cast images on the exposed film. In the cassette interior, the bakelite shall be suitably cushioned with black wool felt; and the bakelite with cushion shall be flush with the frame at the end of the cassette or not more than 1/32 inch below it.

3.3.2.3 Hinges. The cassette shall be provided with a back, hinged along the width and arranged to open a full 180°. The hinges shall be the floating type so that the back shall close parallel to the frame.

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* 3.3.2.4 Back. The back shall be of 0.075 inch thick type 430, No. 4 finish, corrosion-resisting steel with a resilient wool felt cushion arranged to exclude light when the cassette is closed. Narrow slots shall be provided at both ends of the cassette for the projection of the perforated tear-off strip and envelope seal of the film packet. Leaf springs shall be mounted on the corrosion-resisting steel back under the felt cushion to seal the slots and exclude the light. There shall be a 3/16 inch deep by 2 inch wide indentation in the frame to facilitate withdrawal of the tab of the film packet. The hinges shall be retained in two 0.081 inch thick brass, half-hard, satin-chrome plated angles. Each angle shall be secured to the frame by two corrosion-resisting steel countersink head No. 4-40 or 4-36 machine screws, a minimum 1/4 inch long. When closed, the back shall center accurately and shall exert even pressure on the film.

* 3.3.2.5 Springs. When the cassette is closed, firm uniform pressure on the film shall be provided by two heavy leaf springs, mounted by pivot posts riveted to the back and engaging in slots in the frame. Mounted in the center of the back there shall be an eccentric release bar which, when raised approximately 90°, shall release the back sufficiently to permit withdrawal of the film when the cassette is in the processing unit, without restriction of or damage to the film packets. The release bar shall be restrained from being raised more than the 90° position, and shall be so designed that it shall raise the back of the cassette with minimum distortion. The ends of the release bar shall bear on a 21/64 inch diameter type 416 corrosion-resisting steel insert in each side of the frame. The heavy leaf springs shall be 0.0625 inch thick, corrosion-resisting steel, 300 series of FED-STD-66, 3/4 inch wide. The eccentric release bar shall be 0.078 inch thick, type 430 corrosion-resisting steel and shall have a 7/8 inch diameter hole near the top center to facilitate lifting of the release bar. The release bar shall be attached to the cassette back by four hat-shaped brackets constructed of 0.0625 inch thick, type 430 corrosion-resisting steel. Each bracket shall be riveted to the cassette back. Two anodized aluminum strips shall secure the cassette front and the intensifying screen hinges to the cassette frame. Each retaining strip shall be attached to the frame by size No. 4 machine or self-tapping screws. The release bar shall be 10-7/8 $\pm$ 1/32 inches wide. In the raised position, the overall height of the release bar relative to the cassette bottom shall be 1-41/64 $\pm$ 1/64 inch at the ends and 2-9/64 $\pm$ 1/64 inches at the center.

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3.3.2.6 Screen. A 0.031 inch aluminum sheet shall be provided for mounting the intensifying screen. The sheets shall be 10-1/16 + 0, -1/64 inches by 11-1/32 + 1/64 inches and shall be hinged to simplify the loading of the film packet and the cleaning of the screen. Hinges shall float so that the screen will close parallel to the frame. Three small cooper alloy leaf springs shall be equally mounted along each side of the frame to elevate the sheet slightly when the cassette is open. Each leaf spring shall be approximately 5/16 inch wide by 0.005 inch thick and shall extend not more than 3/16 inch into the cassette interior. A 9-5/8 + 0, -1/64 inches by 10-11/16 + 0, -1/64 inches intensifying screen conforming to type I of GG-S-176 shall be securely mounted to the aluminum sheet with the active side facing the film.

* 3.3.3 Carrying case. The carrying case shall consist of an outer trunk-type body and hinged cover. The body section of the carrying case shall be constructed of 3-ply, 3/8 inch plywood panels. The outside shall be covered with 0.020 inch thick hard vulcanized fiber. The fiber covering shall originate at a point on the inside of the plywood box, 2 inches below the top, and shall be securely held on the inside by staples or rivets. The fiber covering shall be securely and smoothly cemented along all four sides and the outside bottom of the box. All joints shall be reinforced along their entire length with hard vulcanized fiber angle strips. The reinforcing angles shall be securely attached with brass tacks of sufficient length to clench flush on the inside of the panels. The eight outside corners of the case shall be protected by approximately 1/32 inch thick metal reinforcements. The reinforcements on the four bottom corners shall be shaped to act as feet which raise the case slightly from the ground. The cover shall be hinged to the body, and shall be made of fiber-covered plywood of the same grade and thickness as the body. The carrying case shall provide space for the processing machine, the repair parts and tools, and three cassettes. A metallic liner shall be riveted within the carrying case, protected with heavy felt for the cassettes. This liner shall also have a space to confine the repair parts and tool kit. Heavy rubber bumpers, shall be riveted to the liner; the liner, in turn, shall be riveted to the body of the chest through additional rubber bumpers. Other rubber bumpers shall be riveted within the cover of the case. These bumbers shall act to confine the processor and prevent damage and abrasion to the finish. A steel clip shall be secured within the cover to accommodate the instruction manuals.

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* 3.3.3.1 Valance. A black enameled valance strip of 0.0375 inch thick steel and 1-1/2 inches wide, shall completely encircle the lower edge of the cover with the ends of the valance meeting under one of the hinges. The valance shall be so positioned on the cover that it shall seat snugly over the metal corner shoulders of the top of the body to make the joint between the cover and the body more secure. The free edge of the valance shall be broken to remove sharp edges.

3.3.3.2 Latches. Two drawbolts shall be firmly affixed to the front of the box, so located that they shall pull the cover down securely on the body when the box is closed. Each drawbolt shall be of rugged construction, and shall be attached by at least two rivets on the valance, by headed tacks clenched on the inside at one or more points on the cover and at three points on the body. Rivet ends shall be headed and smoothly clenched on the interior. The body shall be reinforced in the area above the latch by approximately 1/32 inch thick brass plated steel.

3.3.3.3 Hinges. Cover shall be hinged to the body by three rugged five-knuckle stay hinges. Each hinge shall be attached by at least two rivets on the valance, and by washer backed rivets at a minimum of two points each on the body cover. The degree of stop of the hinge shall be greater than 90° in order to prevent the cover from falling after it has been opened for use.

* 3.3.3.4 Handles. Body shall be equipped with two securely-affixed chest handles one at the center of each end. Handles shall be of steel construction. Handhold of the handle shall be 3-5/8 inches inside measurement, and shall be made of 3/8 inch diameter wire. Handles shall be constructed in such a manner that they shall lie flat when not in use and shall lift 90° from the flat position.

3.4 Tool and repair parts. The class I processing machine shall be supplied with the following tools and repair parts (see 6.9).

	Quantity	Part Name
a.	1	Adapter
b.	1	Tool kit bag
c.	1	3-inch screwdriver
d.	1	Pliers
e.	1	Plastic vial
f.	1	1-1/2 inch screwdriver

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Quantity		Part Name
g.	1	Wiping cloth
h.	1	Feeler gage
i.	2	Hexagonal nut
j.	2	Bearing
k.	2	1/4-inch Tru-Arc ring
l.	1	3/16-inch Tru-Arc ring
m.	2	7/8-inch internal Tru-Arc ring
n.	4	3/8-inch Tru-Arc ring
o.	1	3-ampere fuse
p.	1	Switch
q.	1	No. 8 Allen wrench
r.	1	No. 10 Allen wrench
s.	1	No. 21 Tru-Arc pliers, internal
t.	1	No. 22 Tru-Arc pliers, external
u.	1	5/16 by 13/32 open-end wrench
v.	1	3/8 by 7/16 open-end wrench
w.	1	19/32 by 7/8 open-end wrench
x.	2	8/32 by 3/16 Allen cup-point setscrew
y.	2	10/32 by 3/16 Allen cup-point setscrew
z.	2	No. 10 shakeproof washer, external
aa.	4	2/56 Hexagonal nuts
bb.	2	10/32 Hexagonal nuts
cc.	2	Pressure spring
dd.	2	Needle bearings
ee.	1	Condensor

3.5 Interchangeability of parts. All individual parts, assemblies of parts, or any complete unit supplied under this specification shall be interchangeable with the corresponding parts or assembly of parts of all other complete units supplied under this specification.

3.6 Finish.

3.6.1 Processing machine. The outside of the processing machine and all accessible inside surfaces, except those surfaces which are stainless steel or plated, shall be finished with acid-, alkali-, and solvent-resisting enamel.

3.6.1.1 Degreasing. All surfaces to be enameled shall be thoroughly cleaned and chemically degreased.

* 3.6.1.2 Base primer. After degreasing, a primer coat shall be applied to surfaces to be enameled. The primer shall be zinc-chromate conforming to TT-P-1757.

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3.6.1.3 Secondary primer. This primer shall be applied over the zinc-chromate base primer specified in 3.6.1.2. The primer shall be a synthetic type containing red oxide and other pigments. It shall be baked and shall form a film of excellent adhesion and resistance to moisture, and shall provide a good surface for adhesion of the succeeding enamel coat (see 6.10).

3.6.1.4 Base enamel coat. The primer coats shall be followed by one coat of semigloss green enamel conforming to class B of TT-E-529. The enamel coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking.

3.6.1.5 Spatter coat. One coat of spatter enamel shall be applied over the baked enamel finish specified in 3.6.1.4. The spatter enamel shall be a green, alkyd, semigloss type conforming to class B of TT-E-529, except that it shall be modified by reducing the vehicle to produce a 90 second viscosity through a No. 4 Ford cup at 80°F. The resulting spatter enamel mixture shall be sprayed from a standard spatter gun. The spray application shall be such as to assure a high full texture after baking. The finish shall bake to a hard durable substance without blistering, and shall have excellent abrasion resistance, thoughness, and cleanability.

3.6.1.6 Color. The color of the processing machine shall be color 24087 (see 6.5) of FED-STD-595.

3.6.1.7 Plating and anodizing. All aluminum parts shall be anodized; or nickel or chromium plated. All other metal parts, excepting carrying case hardware and corrosion-resisting steel, shall be cadmium or chromium plated. Carrying case hardware shall be brass plated, with the exception of the side handles which may be cadmium or zinc plated.

3.6.2 Carrying case. A coat of fungus-resistant varnish conforming to MIL-V-173, shall be applied over the vulcanized fiber and exposed areas of plywood of the carrying case.

3.7 Marking.

3.7.1 Processing machine. Each processing machine shall be marked showing the serial number, electrical data, the name or registered trademark of the manufacturer, the letters U.S., the contract or purchase order number, and the National Stock Number. The markings shall be legible and permanent.

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3.7.2 Instructions for use. A pictorial and descriptive instruction sheet shall be securely attached on the underside of the processing machine cover. A copy or sketch of this data shall be forwarded for approval along with the service data (see 3.8).

3.7.3 Wiring diagram. A wiring diagram shall be affixed by means of an adhesive backed mylar overlay to a suitable area of the processor that is visible during use or maintenance. The diagram shall illustrate each electrical component by sketch, outline or symbol; and shall include the name of each component. All wiring and any code used for the identification of the wires shall be shown.

3.7.4 Cassettes. Each cassette shall be permanently and legibly marked with the name or registered trademark of the manufacturer and the type of cassette.

3.7.5 Carrying case. The carrying case shall be permanently and legibly marked with the name or registered trademark of the manufacturer, contract or purchase order number, the letters U.S. and National Stock Number.

* 3.8 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

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* 3.9 Workmanship. Workmanship shall be first class throughout. Construction shall be durable and rugged. All exposed welded parts shall be ground free of scale and protuberances to provide the best possible surface for paint, plating, or anodizing. All exposed parts subject to wear and handling shall have a hard, smooth, durable finish. All bolt connections shall be secured by lockwashers. All members which are secured with setscrews shall be provided with supplementary locking setscrews to prevent loosening by vibration. Unit shall have no sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on processing unit or cassette, as applicable, when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

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4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4, 4.4.1 and 4.4.3.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I.

TABLE I. Sampling for examination.

For visual examination	Inspection level	AQL (percent defective)
Major defects	II	1.0
Total (major and minor defects combined)	II	4.0
For dimensional examination	S-3	2.5

* 4.2.2 For test. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL (precent defective)
Cassette and processing machine	Operation 1/	3.3 thru 3.3.2.6	4.4.1	100%	Zero
Processing machine	Power conformance	3.3.1	4.4.2	2/	Zero
Casette	Opacity of front	3.3.2.2	4.4.3	S-1	6.5

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1/ This test shall be performed to determine functioning of either the cassette or the processing machine. Where both components are being procured concurrently, cassette and processing machine may be tested concurrently.

2/ First article only.

4/3 Examination. The apparatus or its components, as applicable, shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Not complete with all required parts, subassemblies, and materials.
102	Components or parts have cracks, tears, ruptures, or deformed surfaces.
103	Components, mating parts, loose or not properly aligned and fitted.
104	Seam or fillet welds have burn holes or are cracked, fractured, or otherwise not fused.
105	Spot welds not sound.
106	Exposed or accessible edges, corners, and surfaces have burrs, sharpness, or sharp projections.
107	Cassette screen mounting plate back or front not flat; does not rest evenly on a flat surface.
108	Cassette not light tight.
109	Solder or brazing not adherent.
110	Components in the electrical system not in accordance with requirements.
111	Electrical systems and units not consistent in layout and neatly wired.
112	Insulation of wire connectors not clear of, or protected from framework edges to avoid potential of shock hazard or shorts.
113	Electrical wiring has cracks, breaks, or cuts in the insulation.
114	Electrical System has taut leads, loose or improperly fastened terminal connections, or poor solder joints.

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TABLE III. Classification of defects. (cont'd)

Categories	Defects
Minor	
201	Fasteners not tight.
202	Finish has chipped areas, cracks, scratches or roughness.
203	Marking and wiring diagram not present, properly located, legible, correct and permanently affixed.
204	Service data not present in duplicate.
205	Unit not clean.
206	Slag in welds, welds undercut; scale or flux not removed, exhibits pitting, splatter or discoloration
207	Soldering or brazing not clean.

4.3.2 Dimensional examination. The processors shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

* 4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements. Tests for chemical analysis shall be conducted in accordance with method 111 or 112 of FED-TEST METHOD-STD-151.

* 4.4.1 Operation. An unexposed radiographic paper and developer packet shall be loaded in a cassette in a lighted room and processed through the machine in accordance with the manufacturer's directions. An incandescent lamp illuminated by a minimum 100 watts of power shall be positioned 2-1/2 inches from each of the front corners of the processor in a plane perpendicular to the center of the arc circumscribed by the corner radius while the packet is being processed. The machine and cassette shall be observed for conformance to the operating requirements specified in 3.3.2 through 3.3.2.6. The developed film shall be examined for evidence of light leakage, or indication that the developing chemcials did not spread evenly and completely over the image bearing area of the film. Each machine tested shall be operated with either the 10 second or the 45 second type of radiographic paper shown in 6.1.

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* 4.4.2 Power conformance. An exposed radiographic paper and developer packet shall be loaded in a cassette in a lighted room and processed through the machine in accordance with the manufacturer's directions. (The paper shall have been exposed utilizing routine radiographic techniques.) The power applied to the processor shall be 90 volts, 60 Hz. The film shall be examined to determine if normal processing was accomplished. The test shall be repeated at 90 volts, 50 Hz; 120 volts, 60 Hz; and 120 volts, 50 Hz.

* 4.4.3 Opacity of cassette front. A medium speed, high contrast, medical x-ray film, in a cardboard or plastic holder, shall be exposed through the front of the cassette and through an adjacent aluminum 1100 step wedge for 30 milliampere seconds at 60 peak kilovolts at a distance of 30 inches. The film shall be processed in accordance with film manufacturers recommendations and examined to determine the aluminum 1100 equivalent opacity to x-rays and for evidence of voids, imperfections, or included foreign material as specified in 3.3.2.2.

5. PACKAGING

5.1 Preservation packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package.

5.1.1.1.1 Processing machine.

* 5.1.1.1.1.1 Class I. Each processing machine along with accessories shall be positioned, cushioned, and secured within the carrying case (see 3.2.2) in such a manner as to insure against shifting of or damage to the contents during normal handling. The unit shall be further protected by being packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be required.

5.1.1.1.1.2 Class II. Each processing machine shall be packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be required.

5.1.1.1.2 Cassettes. Each cassette shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Class I and class II. Each class I or class II processing machine shall be overboxed in an exterior container designed for a type I load and constructed in accordance with PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Cassettes. Six cassettes shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B. (cassettes only). Six cassettes shall be overboxed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, domestic type or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit and each exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The processing machine and cassette shall be designed to operate with the following items (see 6.8):

6525-00-930-0575	Radiographic Paper and Developer Assembly, 10 Second Processing 9-3/8 by 10 inches, 25s
6525-00-023-4556	Radiographic Paper and Developer Assembly, 45 Second Processing 9-3/8 by 10 inches, 25s

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and included the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. Class required (see 1.2) or cassettes (see 6.4).
d. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 The cassette is listed in the Federal Supply Catalog as follows:

National Stock Number	Item Identification
6525-00-601-0600	CASSETTE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY, 9-3/8 by 10 inches

6.5 Availability of color standards. Requirements for individual color chips (3 x 5 supplements) of the colors referenced in 3.2.2 and 3.6.1.6 should be submitted to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120 citing FED-STD-595 together with the appropriate chip numbers as shown therein.

6.6 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6525-00-823-8127	PROCESSING MACHINE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY, 115 volt, 50-60 Hz, AC, and three cassettes, a carrying case, tools, and repair parts.
6525-00-823-8144	PROCESSING MACHINE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY, 115 volt, 50 Hz, AC, only.

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6.7 Oilite bearings, manufactured by Amplex Division of Chrysler Corporation, are suitable for this application.

6.8 Radiographic Paper and Developer Assemblies are available from Polaroid Corporation, Cambridge, MA or Picker X-Ray Corporation, Medical Products Division, Falls Church, VA.

NSN	Polaroid Catalog No.	Picker Catalog No.
6525-00-930-0575	3000X	192601
6525-00-023-4556	TLX	192607

6.9 Acceptable tools and repair parts as referenced in 3.3 are as follows:

	Picker Part No., or equal
a.	B-18852
b.	B-35383
c.	A-27832
d.	A-27833
e.	A-36254
f.	A-28161
g.	A-28820
h.	A-25208
i.	AT4A-83
j.	T12-80
k.	AT22-23
l.	AT22-38
m.	AT22-39
n.	AT22-22
o.	T27A-13
p.	AT29A-20
q.	T35-7
r.	T35-8
s.	AT75-14
t.	AT75-15
u.	AT75A-1
v.	AT75A-2
w.	AT75A-3
x.	S-33-13
y.	S-39-23
z.	S-93-4
aa.	S-111A-1
bb.	S-111A-27
cc.	T5A-250
dd.	AT12-258
ee.	AT45-25

These tools and spare parts are also available as a complete kit under Picker Catalog number 3295-C from Picker Corporation, 595 Milner Road, Cleveland, OH 44143.

6.10 Examples of secondary primers having the required properties specified in 3.6.1.3 are Finish No. 8798 manufactured by Forbes Finishes or Finish No. 5038 manufactured by Universal Paint and Varnish Company.

* 6.11 A suitable alternate moisture and fungus resistance treatment for that specified in 3.3.1.12 shall be John Dolph Co. Cat. No. AC-29-7S or Allied Finishing Specialities Co. Cat. No. MR-611-F.

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6.12 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6525-1122

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-36148C PROCESSING MACHINE, RADIOGRAPHIC PAPER AND DEVELOPER ASSEMBLY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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